# Year 12 Further Maths - Dimensional analysis

Similar questions - Springs, sails, strings and orbits

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

**1** A mass $m$ is attached to a light spring of stiffness $\kappa$, where $\kappa$ is the force needed per unit extension (units $\mathrm{N\,m^{-1}}$). The mass oscillates, and the time for one complete oscillation is denoted by $\tau$. A student suggests the model $\tau = k m^{\alpha} \kappa^{\beta}$, where $k$ is a dimensionless constant.

**(a)** Find the dimensions of $\kappa$. **[1]**

**(b)** Use dimensional analysis to find $\alpha$ and $\beta$. **[4]**

**(c)** Find the effect on $\tau$ of multiplying $m$ by 4 and $\kappa$ by 9 simultaneously. **[2]**

**(d)** The student finds experimentally that the amplitude of the oscillation has no effect on $\tau$. Explain how this is consistent with the model. **[1]**

**2** The force $F$ exerted by wind on a sail is modelled by $F = k\rho^{\alpha} v^{\beta} A^{\gamma}$, where $\rho$ is the density of air, $v$ is the wind speed relative to the sail, $A$ is the area of the sail and $k$ is a dimensionless constant.

**(a)** Find $\alpha$, $\beta$ and $\gamma$. **[5]**

**(b)** State the effect on $F$ of doubling the wind speed and halving the area. **[2]**

It is suggested that the viscosity $\mu$ of the air should also be included, with dimensions $\mathrm{ML^{-1}T^{-1}}$, so that $F = k\rho^{\alpha} v^{\beta} A^{\gamma} \mu^{\delta}$.

(c) Explain why dimensional analysis alone cannot determine all four indices. **[1]**

(d) Given that $\beta = 1$, find $\alpha$, $\gamma$ and $\delta$. **[4]**

**3** A student models the speed $v$ of a wave travelling along a stretched string by $v = kT^{\alpha} \mu^{\beta} L^{\gamma}$, where $T$ is the tension in the string, $\mu$ is the mass per unit length of the string, $L$ is the length of the string and $k$ is a dimensionless constant.

**(a)** Find the dimensions of $\mu$. **[1]**

**(b)** Determine $\alpha$, $\beta$ and $\gamma$. **[5]**

**(c)** Comment on what the value of $\gamma$ says about the effect of the length of the string. **[1]**

**(d)** Find the factor by which the tension must be multiplied to double the wave speed. **[2]**

**(e)** Another student proposes $v = k\sqrt{TL/\mu}$. Show that this formula is dimensionally inconsistent. **[2]**

**4** A satellite of mass $m$ moves in a circular orbit of radius $r$ about a planet of mass $M$. The universal gravitational constant $G$ has dimensions $\mathrm{M^{-1}L^{3}T^{-2}}$. The period $\tau$ of the orbit is modelled by $\tau = kG^{\alpha} M^{\beta} r^{\gamma}$, where $k$ is a dimensionless constant (the mass $m$ of the satellite is not included).

**(a)** Use dimensional analysis to find $\alpha$, $\beta$ and $\gamma$. **[5]**

**(b)** Find the effect on the period of multiplying the radius of the orbit by 4. **[2]**

**(c)** Explain why, if $m$ were included as well with an unknown index $\delta$, dimensional analysis alone could not determine $\beta$ and $\delta$. **[2]**

**(d)** A student suggests that the escape speed from the planet is $v = k\sqrt{GMr}$. Show by dimensional analysis that this cannot be correct, and write down a correct form. **[3]**

**Question 1**
(a) $[\kappa] = \mathrm{MLT}^{-2}/\mathrm{L} = \mathrm{MT}^{-2}$
(b) $\mathrm{T} = \mathrm{M}^{\alpha}(\mathrm{MT}^{-2})^{\beta}$. Powers of M: $\alpha+\beta=0$. Powers of T: $1=-2\beta$. So $\beta=-\frac{1}{2}$ and $\alpha=\frac{1}{2}$, i.e. $\tau = k\sqrt{m/\kappa}$
(c) $\tau$ is multiplied by $\sqrt{4/9}=\frac{2}{3}$
(d) Amplitude (a length) does not appear in the model, so the model predicts that it has no effect

**Question 2**
(a) $\mathrm{MLT}^{-2} = (\mathrm{ML}^{-3})^{\alpha}(\mathrm{LT}^{-1})^{\beta}(\mathrm{L}^2)^{\gamma}$. M: $\alpha=1$. T: $\beta=2$. L: $1=-3\alpha+\beta+2\gamma=-1+2\gamma$, so $\gamma=1$. $F=k\rho v^2 A$
(b) $F$ is multiplied by $2^2\times\frac{1}{2}=2$
(c) There are only three independent dimensions (M, L, T) giving three equations for four unknowns
(d) T: $-2=-\beta-\delta$, so $\delta=1$. M: $1=\alpha+\delta$, so $\alpha=0$. L: $1=-3\alpha+\beta+2\gamma-\delta=1+2\gamma-1$, so $\gamma=\frac{1}{2}$. So $F=kv\mu\sqrt{A}$

**Question 3**
(a) $[\mu] = \mathrm{ML}^{-1}$
(b) $\mathrm{LT}^{-1} = (\mathrm{MLT}^{-2})^{\alpha}(\mathrm{ML}^{-1})^{\beta}\mathrm{L}^{\gamma}$. T: $-1=-2\alpha$, $\alpha=\frac{1}{2}$. M: $0=\alpha+\beta$, $\beta=-\frac{1}{2}$. L: $1=\alpha-\beta+\gamma=1+\gamma$, so $\gamma=0$
(c) $\gamma=0$: the length of the string has no effect on the wave speed
(d) $v\propto\sqrt{T}$, so $T$ must be multiplied by 4
(e) $[TL/\mu] = \mathrm{MLT}^{-2}\cdot\mathrm{L}\cdot\mathrm{M}^{-1}\mathrm{L} = \mathrm{L}^3\mathrm{T}^{-2}$, so the root has dimensions $\mathrm{L}^{3/2}\mathrm{T}^{-1}\neq\mathrm{LT}^{-1}$

**Question 4**
(a) $\mathrm{T} = (\mathrm{M}^{-1}\mathrm{L}^3\mathrm{T}^{-2})^{\alpha}\mathrm{M}^{\beta}\mathrm{L}^{\gamma}$. T: $1=-2\alpha$, $\alpha=-\frac{1}{2}$. M: $0=-\alpha+\beta$, $\beta=-\frac{1}{2}$. L: $0=3\alpha+\gamma$, $\gamma=\frac{3}{2}$. So $\tau = kr^{3/2}/\sqrt{GM}$
(b) $\tau$ is multiplied by $4^{3/2}=8$
(c) The M equation would become $0=-\alpha+\beta+\delta$ with $\alpha=-\frac{1}{2}$ fixed, so only $\beta+\delta=-\frac{1}{2}$ is determined: two unknowns, one equation. (Physically $\delta=0$.)
(d) $[GMr] = \mathrm{L}^3\mathrm{T}^{-2}\cdot\mathrm{L} = \mathrm{L}^4\mathrm{T}^{-2}$, so $\sqrt{GMr}$ has dimensions $\mathrm{L}^2\mathrm{T}^{-1}\neq\mathrm{LT}^{-1}$. A correct form is $v=k\sqrt{GM/r}$